# Predicate Logic Conti….

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# Properties of Statements:

At least one TRUE Satisfiable

- **Satisfiable:** A statement is satisfiable if there is some interpretation for **which it is true**.
- **Contradiction:** A sentence is contradictory (unsatisfiable) if there is **no interpretation** for **which, it is true**. Always FALSE Contradiction
- **Valid:** A sentence is valid **if it is true for every interpretation**.
- **Equivalence:** Two sentences are equivalent if they have the **same truth value** under every interpretations.

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# Conti...

A statement Q is a logical consequence of P if:
Whenever P is TRUE, Q must also be TRUE.

- **Logical Consequences:** A sentence is a logical consequence of another if it is satisfied by all interpretations which satisfy the first.
- **A valid statement is satisfiable, and a contradictory statement is invalid.**

# Example: On The above definitions:

- **P** is satisfiable but **not valid** since an interpretation **that assigns false to P assigns false to the sentence P**.
- **P V ~P** is **valid** since every interpretation results in a value of **true for (P V ~P ).**
- **P & ~P** is a **contradiction** since every interpretation results in a value of **false for (P & ~P)**.

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# Conti…

- **P and ~(~P)** are **equivalent** since each has the **same truth values** under every interpretation.
- **P** is a **logical consequence** of (P & Q) since any interpretation for which **(P & Q) is true**, **P is also true**.

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# Theorem

- **Theorem 4.1: The sentence s is a logical consequence of $s_1$ , $s_2$ , …… , $s_n$ if and only if $s_1$ & $s_2$ & $s_3$ . . . . & $s_n$ → s is valid.**
- **Proof:** Theorem 4.1 can be seen by first noting that if s is a logical consequence of $s_1$ , $s_2$ , ……. , $s_n$ , then for any interpretation ***I*** in which $s_1$ & $s_2$ & $s_3$ , ….. . &$s_n$ → s is true.
- on the other hand, if $s_1$ & $s_2$ & $s_3$ , …. & $s_n$ →s is valid, then for any interpretation ***I*** if $s_1$ & $s_2$ & $s_3$ , ….. . &$s_n$ is true, **s is also true**.

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# Theorem

- **Theorem 4.2:** The sentence s is a logical consequence of $s_1$ , $s_2$ , ..... , $s_n$ if and only if **$s_1$ & $s_2$ & $s_3$ , .... & $s_n$ & ~s** is **inconsistent**.

- **Proof:** The proof of theorem 4.2 follows directly from **theorem 4.1** since s is a logical consequence of $s_1$ , $s_2$ , ..... , $s_n$ if and only if $s_1$ & $s_2$ & $s_3$ , ..... . &$s_n$ → s is valid, that is, if and only if **~**($s_1$ & $s_2$ & $s_3$ , ..... . &$s_n$ → s ) is inconsistent.

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# Conti .....

- But

$\sim(s_1 \,\&\, s_2 \,\&\, s_3 , \ldots . \,\&s_n \rightarrow s) = \sim(\sim(s_1 \,\&\, s_2 \,\&\, s_3 , \ldots . \,\&s_n) \vee s)$

[**By Conditional Elimination**]

$= \sim\sim(s_1 \,\&\, s_2 \,\&\, s_3 , \ldots . \,\&s_n) \,\&\, \sim s)$

[ **By De Morgan's Law**]

$= s_1 \,\&\, s_2 \,\&\, s_3 , \ldots . \,\&s_n \,\&\, \sim s$

- When s is a logical consequence of $s_1 , s_2 , \ldots , s_n$ , the formula $s_1 \,\&\, s_2 \,\&\, s_3 , \ldots . \,\&s_n \rightarrow s$ is called a theorem, with s is the conclusion.
- When **s** is a logical consequence of the set **S** = $\{s_1 , s_2 , \ldots , s_n\}$ we will also set **S** logically implies or logically entails **S** , written $S \vdash s$ .

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# Table 4.2 lists some of the important laws of PL (Some Equivalence Laws)

| Name of Laws | Statements |
|---|---|
| Idempotency | P V P = P<br>P & P = P |
| Associativity | (PVQ)VR = P V (Q V R)<br>(P & Q) & R = P & (Q & R) |
| Commutativity | P V Q = Q V P<br>P & Q = Q & P<br>P ↔Q = Q↔P |
| Distributivity | P & (Q V R) = (P & Q)V (P &R)<br>P V (Q & R) = (P V Q) & (P V R) |
| De Morgan's Laws | ~(P V Q) = ~P & ~Q<br>~(P & Q) = ~P V ~Q |
| Conditional Elimination | P → Q = ~P V Q |
| Bi-conditional Elimination | P ↔Q = ( p → Q) & (Q →P) |

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# Example

- **Show that P→Q is equivalent to ~PVQ and that P↔Q is equivalent to the expression (P→Q)&(Q→P)**.

- The truth table 4.3 is given bellow.

# TABLE 4.3 : Truth table for equitant sentences

| P | Q | ~P | (~PVQ) | (P→Q) | (Q→P) | (P→Q)& (Q→P) |
|---|---|---|---|---|---|---|
| T | T | F | T | T | T | T |
| T | F | F | F | F | T | F |
| F | T | T | T | T | F | F |
| F | F | T | T | T | T | T |

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# Inference Rules

- **The inference rules of PL provide the means to perform logical proofs or deductions.**

- Few Such Rules are as follows:
  - **Modus ponens**
  - **Chain Rule**

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# Modus Ponens:

- From P and P → Q infer Q. This sometimes written as
- P
- P → Q
- ---
- Q

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# Example For Modus Ponens:

- Given: (Joe is a father)
- And: (Joe is a father) → ( Joe has a child)
- Conclude: (Joe has a child)

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# Chain Rule

- Form $P \rightarrow Q$ and $Q \rightarrow R$, infer $P \rightarrow R$.
- Or
- $$\frac{P \rightarrow Q \quad Q \rightarrow R}{P \rightarrow R}$$

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# Example for Chain Rule

- Given: (programmer likes LISP)→ (programmer hates COBOL)
- **and** : (programmer hates COBOL) → (programmer likes recursion)
- **Conclude**: (programmer likes LISP)→ (programmer likes recursion)

- LISP → **Lis**t **P**rocessing
- COBOL → **Co**mmon **B**usiness **O**riented **L**anguage
- Prolog → **Programming in Log**ic

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# Assignment-3

1. Construct a **truth Table** for the expression **(A & (A V B))**.
2. Determine whether each of the following sentences is
   (a) Satisfiable
   (b) Contradictory
   (c) Valid

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# Assignment-3 Conti…

- (i) $S_1$: (P & Q) V ˜(P & Q)
- (ii) $S_2$: (P V Q )→ (P & Q)
- (iii) $S_3$: (P V Q)→ R V ˜Q
- (iv) $S_4$: (P V Q)& (PV ˜Q) V P
- (v) $S_5$: P → Q→ ˜P
- (vi) $S_6$: P V Q & ˜P V ˜Q &P

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- **Thanks**

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